

Clinical NLP for registries and computational phenotyping

Deploy the existing abstractor. Evaluate with MLflow.
Build the evidence for Amber.

From the D3 vision to a concrete first use case

D3 UPDATE · SLIDE 17

Registry & computational phenotyping tools

Use NLP and AI to reduce the time required to build phenotypes for registries and pragmatic trials.

D3 UPDATE · SLIDE 18

MLflow for model governance

Use the Databricks environment for reproducible model versioning, evaluation, and monitoring.

Our starting point: extract usable oncology variables from clinical notes, then evaluate them against expert review.

Note abstraction supplies inputs to phenotyping; patient-level phenotype logic remains a separate step.

An existing oncology abstractor is the foundation

A Python implementation of the Rails **omop-abstractor** workflow.

CLINICAL CONTENT

Structured fields

Histology, site, laterality, procedure, grade, stage, and receptor status.

ONCOLOGY CONTEXT

Configured tasks

Note selection, local and outside reports, specimen groups, and cohort vocabularies.

RESEARCH OUTPUTS

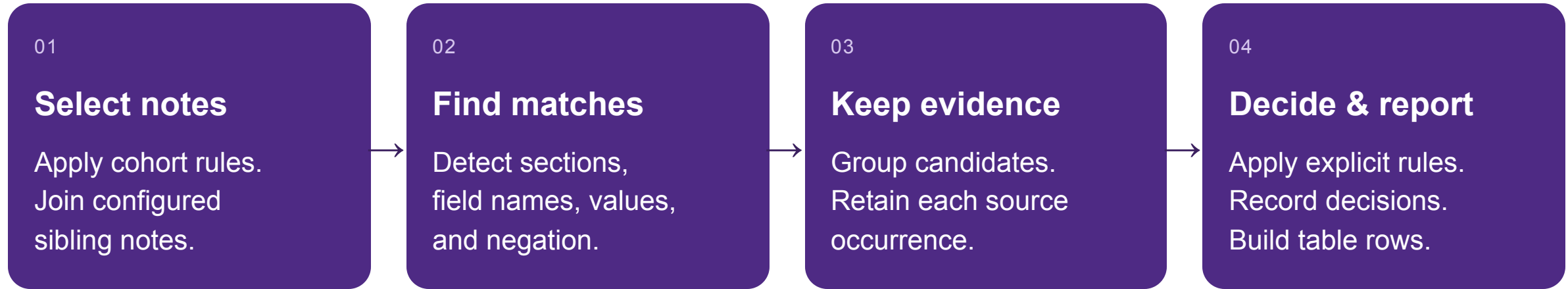
Reviewable tables

Report rows with candidate values, source matches, and decision reasons available for audit.

Suites include breast, lung, colorectal, AML, lymphoma, and other oncology cohorts.

Preserve useful clinical rules and reporting conventions while moving execution into Python.

Collect the evidence. Then choose the value.



Evidence stays inspectable

A match records its text, section, location, and negation status—even when it loses.

Uncertainty stays visible

Fields can be valued, ambiguous, unknown, not applicable, or abstained after an LLM call.

A final value is accompanied by the candidates considered and the reason for the decision.

Use LLMs where interpretation adds value

SYNTHETIC ILLUSTRATION

DCIS: nuclear grade 3.

Invasive carcinoma:
Nottingham grade 2.

Finding both grades is only the first step.
Each grade must belong to the correct field.

Optional resolution after NLP

- Resolve competing candidate values.
- Check values shared across related fields.
- Extract related fields together, such as ER, PR, and HER2.
- Validate answers and retain the original evidence.

Compare rules alone with rules + LLM resolution on the same cases.

Deploying the abstractor into the Databricks EDW

We are deploying the existing NLP abstractor tool into the **Databricks Enterprise Data Warehouse environment**.

DATABRICKS EDW · DEPLOYMENT WORKFLOW

EDW notes

Selected cohort
and source metadata



Python abstractor

Existing schemas
and optional LLM resolution



Research tables

Structured variables
with review and audit views

Build on the existing environment

Databricks notebooks connect source queries, abstraction, and report generation.

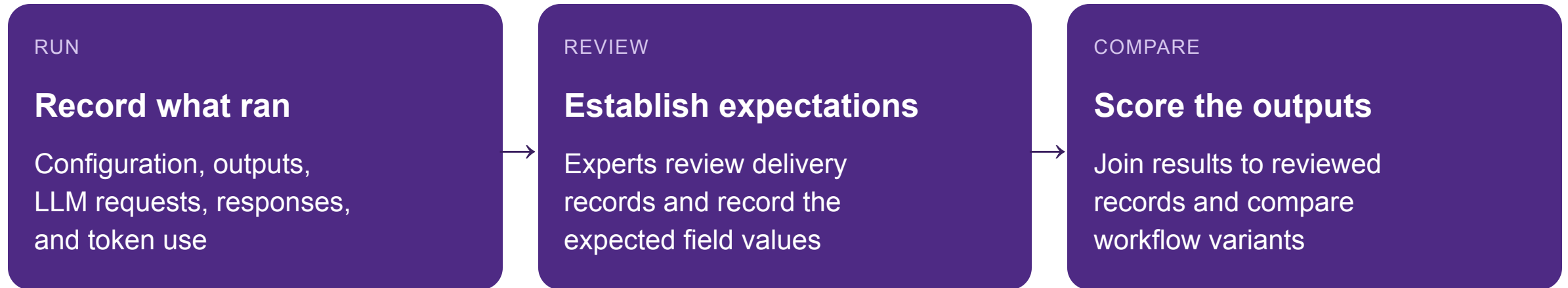
Clinical text, model requests, and review artifacts stay in institutionally approved destinations.

Evaluate the deployed workflow

MLflow links runs and LLM traces to expert-reviewed evaluation records.

Using MLflow to evaluate results

We are using **MLflow in Databricks** to connect extraction results, expert expectations, and reproducible comparisons.



Current comparison

Deterministic abstraction versus abstraction with selected LLM resolution.

Foundation for Amber

Reuse the evaluation approach and compatible task definitions for the next generation.

Successful execution is only the start; expert-reviewed outputs determine quality.

Let the baseline identify the next investment

ADAPTATION

A new task

How much expert work goes into field definitions, vocabularies, rules, and prompts?

TRUST

A defensible answer

Can a reviewer trace a value to text that actually supports the intended clinical meaning?

EFFORT

A useful workflow

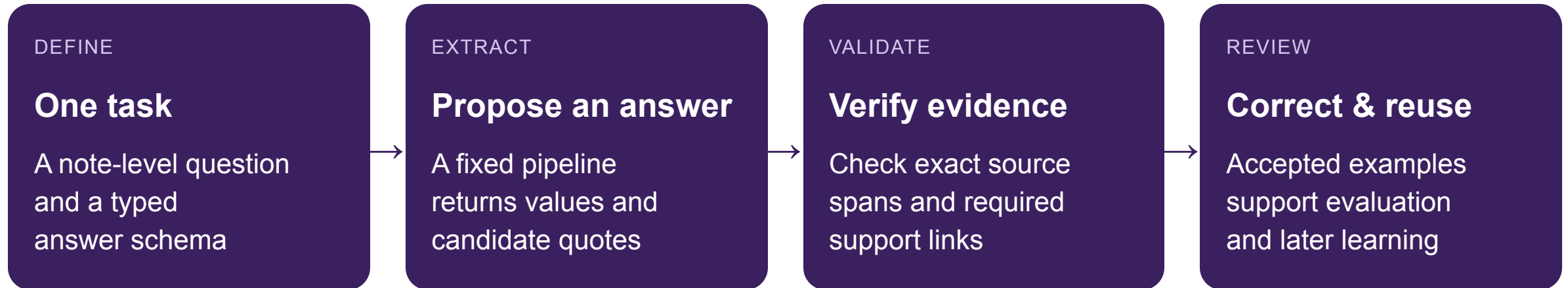
Does assisted review reduce total expert time at an acceptable quality level?

These questions shape Amber's design and evaluation.

Design goals are hypotheses. Improvements require a measured comparison.

Amber: a proposed next generation

Structured clinical answers with **verified source evidence** and a reusable correction workflow.



The goal: less total expert effort at a defined clinical quality target.

Under development. The full extraction and correction workflow is planned.
Agents are a later experiment; source validation alone does not establish clinical correctness.

What changes—and what carries forward

Dimension	Existing abstractor	Amber: proposed direction
Starting point	Implemented oncology workflows; Databricks EDW deployment underway	Early development; start with one measured clinical task
Extraction	Configured NLP rules + optional LLM resolution	Typed task + fixed extraction pipeline; test agents only when justified
Evidence	Source matches, candidates, and explicit decision reasons	Validate source spans and the full chain of support before saving claims
Review & reuse	Delivery records, expert expectations, and MLflow evaluation	Canonical corrected examples for evaluation and later learning
Platform	Databricks integration and legacy-compatible report outputs	Portable core and open-source MLflow APIs; Databricks by configuration

Carry forward clinical knowledge; measure quality, effort, and cost.

Compare on the same task, cases, and quality target

One shared protocol

1. Choose an overlapping note-level task and define acceptable answers.
2. Set quality targets before tuning; split by patient or document.
3. Compare rules, rules + LLM, and Amber's fixed pipeline when ready.
4. Include manual abstraction as the reference for expert effort.

Report more than accuracy

Correctness · right values and missed answers

Evidence · valid locations and clinical support

Coverage · automatic completion, abstentions, failures

Effort & cost · setup, review, correction, latency, model use

Track comparisons in MLflow; use expert judgment to assess clinical support.

Amber's first planned dataset: CORAL v1.0, with 40 expert-labeled notes.

A pilot, not a broad validation claim; a shared task requires validated label mapping.

Brim and Amber can serve complementary roles

Brim Analytics is undergoing a trial at Northwestern.

The pilot can inform how researcher tools and institutional infrastructure fit together.

BRIM PILOT · RESEARCHER WORKFLOW

Author and review variables

- Define study variables.
- Review extracted values and evidence.
- Produce project-level data.

AMBER · PROPOSED INFRASTRUCTURE

Make the evidence reusable

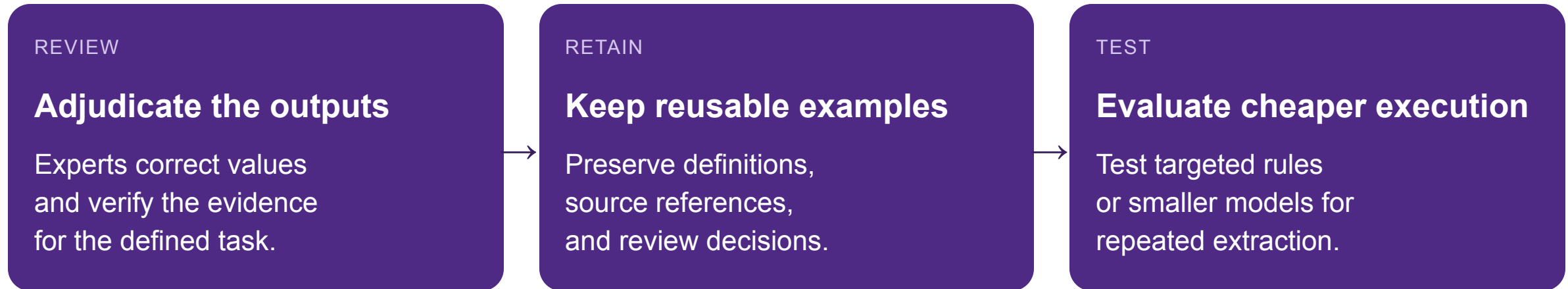
- Validate source-linked claims.
- Version extraction and evaluation.
- Retain evidence for EDW reuse.

A successful pilot could inform both product adoption and institutional infrastructure.

Proposed relationship; interoperability remains to be assessed. [Brim Analytics](#)

Turn expert-reviewed cases into reusable assets

The pilot could produce more than a study table:
reviewed values, variable definitions, and supporting source spans.



Hypothesis: reviewed examples can lower recurring cost at matched clinical quality.

Reuse requires permitted exports and verified evidence. Evaluate on an independent expert-labeled test set.

What we should learn from the Brim pilot

Use the pilot to assess **clinical value, operational fit, and reuse.**

QUALITY & EFFORT

Does it help experts?

Measure correct values, omissions, evidence support, and time to define and review variables.

PORTABILITY & REUSE

What can we retain?

Confirm export of definitions, values, source locations, review history, and run configuration.

DEPLOYMENT & COST

Where does it fit?

Assess EDW integration, permitted endpoints, and total cost at study and recurring-cohort scale.

Use MLflow for the common evaluation; let measured results guide adoption and development.

Comparison with the existing abstractor—and Amber when ready—requires aligned tasks and independent expert review.

Deliver the baseline. Build the comparison.

NOW · DEPLOY & EVALUATE

Existing abstractor

Deploy in the Databricks EDW.
Evaluate with MLflow.
Review errors with experts.

NEXT · BUILD & TEST

Amber's first task

Define acceptance criteria.
Validate source evidence.
Add minimal correction.

THEN · COMPARE & DECIDE

Evidence for adoption

Compare matched cases.
Measure total expert effort.
Expand when results justify it.

A practical path toward D3's registry
and computational phenotyping tools.